

Project plan

Advantech Demo Box Project Integration

Advantech Eindhoven

Client: Advantech Europe B.V.

Location: Science Park Eindhoven 5708, 5692 ER Son, The Netherlands

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Communication

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V1	27/02/2025	Company supervisor and mentor
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1. Project Assignment

1.1 Context

Advantech Europe B.V. specializes in **industrial IoT, embedded systems, and automation solutions**. The company is developing a **Demo Box**, an interactive system showcasing its industrial hardware and software capabilities.

The **Demo Box** integrates Advantech's devices, such as:

- **EKI Series Industrial Switches** – Handle network connectivity and data transfer.
- **ADAM Remote I/O Modules** – Acquire and process sensor data.
- **Panel PC (Touch Display)** – Provides real-time system control and visualization.



Photo of the **Demo Box**, that me and my team are going to be working on.

These devices need **full integration** with **WISE-PaaS**, Advantech's **IoT cloud platform** for **remote monitoring, data visualization, and predictive maintenance**.

The demo box is partially built, but lacks full integration between its hardware components and software capabilities for remote monitoring. My internship will focus on completing this integration to create a fully functional, interactive demo kit.

1.2 Goal of the project

The goal of this project is to:

- Establish seamless communication between Advantech's devices within the demo box.
- Integrate WISE-PaaS for remote monitoring and device management.
- Develop a user-friendly front-end interface to display real-time data.

By achieving this, the demo box will serve as a powerful sales and engineering tool to showcase Advantech's IoT solutions to customers.

1.3 The assignment

The primary task is to fully integrate Advantech's hardware and software within the demo box, ensuring seamless communication and remote monitoring.

Key Deliverables:

- Device Integration: Ensure communication between the EKI switch, ADAM modules, and panel PC.
- WISE-PaaS Integration: Enable remote monitoring and data visualization.
- UI Development: Build a user-friendly interface for interaction with the demo system.
- Testing & Debugging: Ensure the setup functions correctly and efficiently.

1.4 Scope

<< Indicate the scope. If necessary, make a context diagram for clarification that shows the relationships with other systems and the environment. This section should also describe what will not be delivered. For example, if you agree to deliver a high fidelity prototype, then (part of) the implementation and management falls outside your scope. Make this as concrete as possible so that there are no misunderstandings between you and your client.

>>.

The project includes:	The project does not include:
1 Integration of hardware devices in the demo box.	1 Deployment outside of the demo environment .
2 Development of WISE-PaaS-based remote monitoring .	2 Full commercial software development beyond the demo use case.
3 Implementation of a UI for real-time data visualization .	3 Large-scale production or hardware modifications.
4 Testing, debugging, and documentation of the system.	

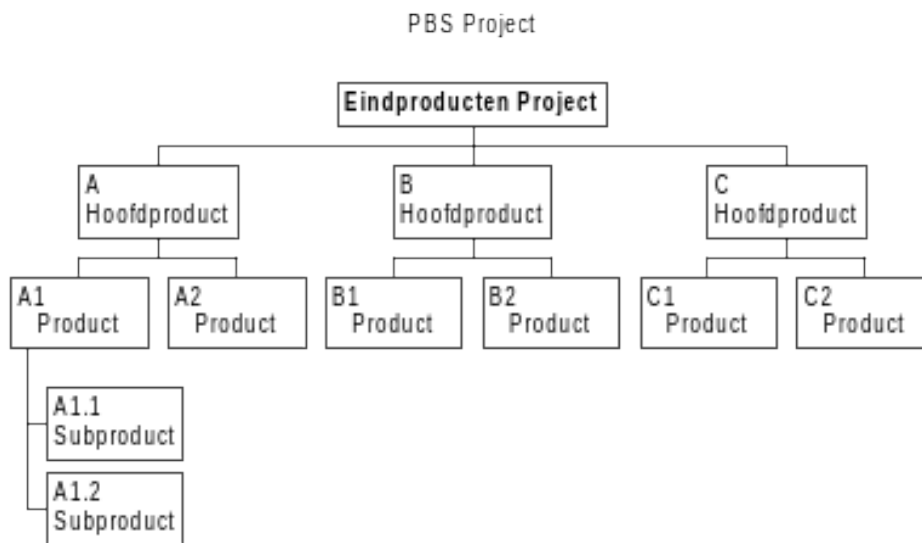
1.5 Conditions

1. **All development** must be compatible with **Advantech's existing infrastructure**.
2. **WISE-PaaS** must be the primary platform for remote monitoring.
3. The project must be completed **within the internship period**.

1.6 Finished products

<<A Product Breakdown Structure of the end and intermediate products that the project will deliver with a short description in text of each product. The end products are more than the project plan and the product itself. Also, for example, requirements and architecture documents and research and test reports are typical parts of a PBS. These documents are important for the relevant stakeholders during development as well as during the transfer and during the management phase. During the project you can change the PBS and you can add or remove products in consultation. >>

Deliverable	Description
Integrated Demo Box	A fully functional system with real-time data exchange.
WISE-PaaS Remote Monitoring	Enables cloud-based device management.
Interactive UI	Dashboard to display sensor data, control devices, and analyze performance.
Technical Documentation	Manuals detailing integration, setup, and troubleshooting.



1.7 Research questions

1. Main Research Question:
How can Advantech's hardware and software solutions be integrated into the demo box to demonstrate seamless communication, remote monitoring, and device management?
2. Sub-Research Questions:
 1. What are the best practices for enabling communication between devices?
 2. How can WISE-PaaS be leveraged for remote monitoring?
 3. What UI strategies work best within WISE-PaaS?

2. Approach and Planning

2.1 Approach

The project will follow a **Scrum-based Agile approach** with **bi-weekly sprints** to ensure iterative development and feedback loops.

2.1.1 Test approach

2.2 Research methods

Research Question	Methodology
How can Advantech WISE-PaaS be integrated with the hardware setup?	Hands-on testing, reviewing Advantech documentation, consulting engineers.
What are the UI restrictions and possibilities of WISE-PaaS?	Exploring platform documentation, UI prototyping, usability testing.
What strategies ensure reliable communication between Advantech devices?	Network testing, protocol analysis (Modbus, MQTT), troubleshooting existing issues.

2.3 Learning outcomes

<<Discuss how you are going to demonstrate each learning outcome in the project. The easiest way is to think about which of the professional products you are going to use as evidence for each of the learning outcomes.>>.

2.4 Breakdown of the project

Phase	Task	Deliverables
Phase 1	Research & training	Research documentation
Phase 2	Hardware communication setup	Working communication layer
Phase 3	WISE-PaaS integration	Remote monitoring functionality
Phase 4	UI development	Front-end dashboard
Phase 4	Testing & debugging	Final working demo box

2.5 Time plan

Methodology:

- 1.Sprint 1 → Research & technical training
- 2.Sprint 2 → Hardware communication setup
- 3.Sprint 3 → WISE-PaaS integration
- 4.Sprint 4 → UI development
- 5.Sprint 5 → System testing & debugging
- 6.Sprint 6 → Final improvements & documentation

Phasing	Effort	Start	Ready
1			
2			
3			

3. Project Organization

3.1 Team members

<< Describe the organization of the project with its immediate environment. An organization chart can be displayed for clarification. Indicate in descriptive form which roles are included in the organization chart with the associated authorities and responsibilities. It must be clear who is authorized to do what and what can be expected of whom. Indicate who is involved in your project and what his/her function is and what the role is within your project. For example, someone with the function 'manager of department X' can have the role of Product Owner in your project. In this project, both the internship/graduation organization and Fontys are stakeholders. So include your internship teachers and yourself in this schedule.>>

Name	Abbr.	Role/tasks	Availability
Georgi Balabanov	Sometimes it is useful to indicate a role or person with an abbreviation	Intern	Full-time
Advantech mentor		Mentor/Team Leader	Weekly check-ins
Advantech Engineers		Technical Support	On-demand

3.2 Communication

1. **Weekly meetings** with the mentor.
2. **Bi-weekly sprint reviews** with stakeholders.
3. **Slack/email** for daily updates.

3.3 Test environment

The test environment includes:

- **Physical Demo Box:**
 - **EKI Series Industrial Switch** (handles communication).
 - **ADAM Remote I/O Modules** (data acquisition).
 - **Panel PC** (user control & display).
- **WISE-PaaS cloud environment** for software testing.
- **Simulated industrial network** for integration testing.

Testing will be performed in a controlled lab setup.

3.4 Configuration management

Version Control System: Git (GitHub or internal Advantech repo).

1 . Branching Strategy:

- **Main Branch:** Stable versions of the system.
- **Development Branch:** Active feature development.

- **Feature Branches:** Separate work areas for UI, backend, and integration.

Releases:

- **Alpha Release:** Internal testing of early integration.
- **Beta Release:** Functionality tests with stakeholders.
- **Final Release:** Fully tested Demo Box.

4. Finance and Risks

4.1 Cost budget

No major costs expected, as all necessary hardware and software are provided by Advantech.

4.2 Risks and fall-back activities

<<Define risks. What have you already included in the plan to limit or prevent the risk? What choice is made if the risk does unexpectedly occur? Think of organizational risks (such as the sudden leave of the company supervisor) as well as more substantive risks (for example, what to do if you find out during your internship that it is better for the company to purchase an external application instead of the application to be developed).

Think of real risks that can actually influence your project. For example, there may be a risk that your company supervisor will be absent, for example due to illness or because he is going to do something else. Is there a backup in the company?
>>

Risk	Prevention activities included in plan	Fall-back Activities
1 Failing Technical Training	Extra mentor support, online learning	Additional training sessions
2 UI Limitations in WISE-PaaS	Early testing & prototyping	Adjust design based on feasibility
3 Delays in Integration	Regular progress tracking	Extend sprint if necessary

5. Other

- Exploring the feasibility of **multilingual support** in **WISE-PaaS UI**.
- Ensuring **accessibility and user experience** in the final product.

5.1 Detailed Scope

- Integration includes EKI-2525 switch, ADAM-6050 & ADAM-6017 modules, and PPC-3120 panel PC.
- WISE-PaaS platform version 4.0 for cloud integration.
- Scope diagram provided illustrating hardware/software relationships clearly.
- Excludes deployment to production or external client environments.
- No hardware redesign or modification beyond basic integration.

5.2 Expanded Finished Products (Product Breakdown Structure)

- Integrated Demo Box: Hardware setup guide, Protocol documentation (MQTT, Modbus), Integration report.
- WISE-PaaS Remote Monitoring: Integration report, WISE-PaaS setup and configuration manual.
- Interactive UI: Wireframes/mockups, Usability testing report, Final UI implementation documentation.
- Technical Documentation: User manuals, Troubleshooting guides, Maintenance and future integration instructions.

5.3 Explicit Project Organization

- Georgi Balabanov (Intern): Responsible for research, hardware/software integration, UI development, documentation.
- Mentor / Team Leader: Guidance, technical consultation, review of progress.
- Advantech Engineers: Provide technical support, resolve specific hardware/software integration challenges.
- Fontys Internship Coordinator: Regular progress evaluations, educational oversight.

5.4 Learning Outcomes Clarification

- Integration tasks demonstrate ability to work with IoT technology and cloud platforms.
- UI development highlights competency in user-centered design principles.
- Project documentation will evidence professional-level communication and reporting skills.
- Sprint reviews and project management reflect agile project management skills.

5.5 Detailed Risks and Fallback Activities

- Risk: Hardware integration failure - Prevention: early-stage prototyping; Fallback: alternate compatible hardware.
- Risk: Delays in cloud integration - Prevention: regular meetings with cloud engineers; Fallback: extend integration phase.
- Risk: UI limitations in WISE-PaaS - Prevention: early prototyping; Fallback: scale back UI complexity or find workaround solutions.